NPTEL ETHICAL HACKING & NETWORK COMMUNICATIONS

WEEK 6 DETAILED CLASSROOM STUDY NOTES & EXAM PREPARATION NOTEBOOK

Course: NPTEL Ethical Hacking | Coverage: Lectures 26 to 30 (Basic Cryptography, Private-Key Ciphers, DES/AES, RSA & Diffie-Hellman)

MODULE 1: LECTURE 26 — BASIC CONCEPTS OF CRYPTOGRAPHY & SECURITY ATTACKS

1.1 Introduction to Cryptography in Cybersecurity

1. Simple Explanation: Imagine you want to send a secret letter to your friend across a crowded classroom. If you write it in plain English, anyone who intercepts the paper can read your secrets. To prevent this, you scramble the letters using a secret code that only you and your friend understand. In cybersecurity, **Cryptography** is the science and mathematical practice of scrambling digital messages so that adversaries cannot read or alter them.

2. Why it is Needed: Hacking involves identifying and exploiting vulnerabilities in systems, networks, or infrastructure to gain unauthorized access or cause disruption. While it is impossible to eliminate all software bugs or security flaws, cryptography provides an essential defense mechanism that protects sensitive data even when the underlying network channel is insecure.

3. Technical Definition: *Cryptography* is the discipline concerned with algorithmic techniques for securing communication and information against adversarial activity, providing confidentiality, data integrity, authentication, and non-repudiation.

> **REMEMBER: KERCKHOFFS'S PRINCIPLE**
>
> **Core Principle of Security:** Security must reside entirely in the secrecy of the *key*, not in keeping the encryption algorithm secret (Kerckhoffs's Principle). An intruder can know the exact algorithm being used, but without the secret key, decrypting the ciphertext remains computationally infeasible.

1.2 Classification of Security Attacks

A **Security Attack** is any action that compromises the security of information belonging to an individual or organization. Attacks are broadly categorized by the security property they target:

Attack Category	Targeted Security Property	How It Works (Mechanics)	Real-World / Lecture Example
Interruption	Availability	An intruder prevents legitimate users from accessing a system, service, or communication link (e.g., dropping packets at a router).	Denial of Service (DoS) attack flooding a web or email server so legitimate users cannot connect.
Interception	Confidentiality	An unauthorized third party gains silent access to data during transmission without modifying it.	Sniffing confidential banking details or personal messages flowing over an unencrypted network link.
Modification	Integrity	An intruder intercepts a message, alters its contents, recalculates necessary checksums, and forwards the modified data.	Hacking a military or financial channel to change order amounts or strategic commands in transit.

Attack Category	Targeted Security Property	How It Works (Mechanics)	Real-World / Lecture Example
Fabrication	Authenticity	An intruder injects fake data or fabricated messages into the network, pretending to be a legitimate user.	Sending phishing emails or admin commands disguised as coming from a trusted bank or network administrator.

1.3 Passive vs. Active Attacks

Attacks are further categorized by their operational behavior into **Passive** and **Active** attacks:

- **Passive Attacks:** The intruder monitors, eavesdrops, or analyzes traffic without altering system resources or transmitted data. Because no data is altered and no extra traffic is injected, passive attacks leave no traces and are *extremely difficult to detect*. Examples include message content reading and traffic analysis (analyzing communication patterns, frequency, and message volume to infer network structures). Prevention relies entirely on strong encryption.
- **Active Attacks:** The intruder actively modifies data, injects false packets, replays captured sessions, or disrupts services. Active attacks alter system state and cause noticeable disruptions, making them *easier to detect* but harder to prevent in real-time. Types include Masquerade, Replay, Modification, and Denial of Service (DoS).

1.4 Essential Security Services

To counter these security attacks, robust system architectures implement six fundamental **Security Services**:

1. **Confidentiality:** Ensures information is kept private and protected against unauthorized disclosure or eavesdropping (achieved via Encryption).
2. **Authentication:** Verifies the true identity of a user, system, or the genuine origin of a message (prevents Masquerading and Fabrication).
3. **Integrity:** Guarantees that data has not been altered, corrupted, or tampered with during storage or transmission (achieved via Hashes and MACs).
4. **Non-Repudiation:** Prevents a sender or receiver from falsely denying their participation in a transaction or message transmission (achieved via Digital Signatures).
5. **Access Control:** Regulates and restricts who can access specific system resources, memory, or administrative permissions.
6. **Availability:** Guarantees that authorized users have timely, uninterrupted access to essential systems and services (prevents DoS/Interruption).

EXAM POINT: ATTACK CLASSIFICATIONS TO MEMORIZE

Exam Focus (NPTEL Assignment 6 Question):
- *Question:* A DoS attack that makes a system or service inaccessible to legitimate users falls under which attack category?
- *Answer:* **Interruption** (an attack on Availability).
- *Question:* An attack where an attacker pretends to be a legitimate entity to gain unauthorized access is called?
- *Answer:* **Masquerade Attack** (an Active Attack on Authenticity).

MODULE 2: LECTURE 27 — PRIVATE-KEY CRYPTOGRAPHY & CLASSICAL CIPHERS

2.1 Principles of Private-Key (Symmetric) Cryptography

1. Simple Explanation: Private-Key (Symmetric) cryptography works like a traditional physical padlock. The sender locks (encrypts) a box using a specific physical key, and the recipient opens (decrypts) the box using an *exact copy of that same key*.

2. Mathematical Formulation:

- **Plaintext (P):** The original, readable message or digital data.
- **Ciphertext (C):** The scrambled, unreadable output generated after encryption.
- **Shared Secret Key (K):** The secret value known only to the sender and recipient.
- **Encryption Algorithm:** $C = E(P, K)$
- **Decryption Algorithm:** $P = D(C, K)$

2.2 The Symmetric Key Distribution Bottleneck Formula

A critical operational drawback of symmetric cryptography is **Key Management**. If N distinct parties need to communicate securely in pairs, every unique pair must share a distinct secret key.

FORMULA SHEET & WORKED EXAMPLE: SYMMETRIC KEY COUNTING

Symmetric Key Requirement Formula:
$$\text{Total Keys Required} = \binom{N}{2} = \frac{N(N - 1)}{2}$$
• **Symbol Explanation:** N = Number of communicating parties in the network.
• **Numerical Example (NPTEL Assignment 6, Q10):** If $N = 40$ parties want to communicate securely using symmetric encryption:
$$\text{Total Keys} = \frac{40 \times (40 - 1)}{2} = \frac{40 \times 39}{2} = \frac{1560}{2} = \mathbf{780 \text{ distinct keys.}}$$
• **Scalability Impact:** For $N = 1,000$ parties, total keys $= 1000 \times 999 / 2 = 499,500 \approx 500,000$ keys! Maintaining key secrecy for half a million keys becomes impossible without public-key infrastructure.

2.3 Classical Substitution Ciphers

1. Caesar Cipher (Shift Cipher): Each plaintext letter is replaced by shifting it k positions down the alphabet (wrapping around $Z \to A$).

- *Mathematical Shift Formulas ($A=1, B=2, \dots, Z=26$):*
  $$\text{Encryption: } C = (P + k - 1) \bmod 26 + 1$$
  $$\text{Decryption: } P = (C - k + 25) \bmod 26 + 1$$
- *Vulnerability:* Key space is extremely small (only 25 possible keys, $k=1 \dots 25$). An attacker easily breaks it via **Brute Force** (testing all 25 shifts).

SOLVED LECTURE NUMERICALS: CAESAR CIPHER ENCRYPTION & DECRYPTION

1. Encryption Numerical (NPTEL Assignment 6, Q2): Encrypt plaintext 'SECURE' with Caesar shift $k = 3$.
• $S (19) \to (19+3-1)\%26+1 = 22 \implies \mathbf{V}$
• $E (5) \to (5+3-1)\%26+1 = 7 \implies \mathbf{H}$
• $C (3) \to (3+3-1)\%26+1 = 5 \implies \mathbf{F}$
• $U (21) \to (21+3-1)\%26+1 = 23 \implies \mathbf{X}$
• $R (18) \to (18+3-1)\%26+1 = 20 \implies \mathbf{U}$
• $E (5) \to 7 \implies \mathbf{H}$ $\implies$ Final Ciphertext: **'VHFXUH'**.

2. Decryption Numerical (NPTEL Assignment 6, Q5): Decrypt ciphertext 'KHOOR' with Caesar shift $k = 3$.
• $K (11) \to K - 3 = 8 \implies \mathbf{H}$
• $H (8) \to H - 3 = 5 \implies \mathbf{E}$
• $O (15) \to O - 3 = 12 \implies \mathbf{L}$
• $O (15) \to 12 \implies \mathbf{L}$
• $R (18) \to R - 3 = 15 \implies \mathbf{O}$ $\implies$ Final Plaintext: **'HELLO'**.

2. Mono-alphabetic Cipher: Replaces each plaintext letter with an arbitrary mapped letter according to a fixed 26-letter permutation key.

- *Key Space:* $26! \approx 4.03 \times 10^{26}$ possible keys. Brute force is computationally impossible.
- *Vulnerability:* Broken quickly using **Frequency Analysis**. In natural languages (e.g., English), individual letters (E, T, A) and letter pairs (TH, HE) occur with predictable frequencies. Ciphertext letter frequencies match plaintext letter frequencies directly.
- *Lecture Example (NPTEL Assignment 6, Q3):* Given mapping key (A B C D E ...) mapped to (Q W E R T ...):
  To encrypt 'CODE': $C \to E$, $O \to G$, $D \to R$, $E \to T \implies \mathbf{'EGRT'}$.

2.4 Transposition Ciphers & Stream vs. Block Architecture

• **Transposition Ciphers:** Do not alter the letters themselves; instead, they rearrange (permute) character positions (e.g., writing text into 7 columns row-by-row and reading out column-by-column). Letter frequencies remain completely unchanged, making pattern analysis simple.

• **Stream Ciphers:** Encrypt data continuously bit-by-bit or byte-by-byte in real time using a generated keystream. Ideal for live streaming audio/video where low latency is required.

• **Block Ciphers:** Process plaintext in fixed-size blocks (e.g., 64-bit or 128-bit blocks) at a time. If the message length is not a multiple of the block size, padding bits are added to fill the final block.

MODULE 3: LECTURE 28 — EVOLUTION OF PRIVATE-KEY ALGORITHMS: DES, 3DES & AES

3.1 Data Encryption Standard (DES) Architecture

1. Overview: The Data Encryption Standard (DES) was a pioneering symmetric block cipher adopted in the 1980s. It processes data in **64-bit plaintext blocks** and uses a **56-bit effective key** (stored as 64 bits including 8 parity bits).

2. Feistel Cipher Structure Mechanics:

- *Initial Permutation (IP):* The 64-bit block undergoes a fixed transposition of bit positions.
- *16 Processing Rounds:* The permuted block is split into two 32-bit halves: Left (L) and Right (R).

- *Round Execution Formula:* In each round i (from $i=1 \dots 16$):
  $$L_i = R_{i-1}$$
  $$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$
  where F is a non-linear round function and K_i is a 48-bit subkey derived from the 56-bit master key.
- *Inverse Permutation (IP^{-1}):* After 16 rounds and a 32-bit swap, the final transposition produces the 64-bit ciphertext.

3. Fatal Security Defect: A 56-bit key space contains $2^{56} \approx 7.2 \times 10^{16}$ keys. Modern supercomputers break DES in hours using exhaustive **Brute Force** key search.

3.2 Triple DES (3DES)

To address DES's weak key length without replacing existing DES hardware, 3DES applies single DES three times per block in **Encrypt-Decrypt-Encrypt (EDE)** mode using three distinct 56-bit keys (K_1, K_2, K_3):

$$\text{Encryption: } C = E_{K3}\left(D_{K2}\left(E_{K1}(P) \right) \right)$$

$$\text{Decryption: } P = D_{K1}\left(E_{K2}\left(D_{K3}(C) \right) \right)$$

- **Effective Key Size:** $3 \times 56 = \mathbf{168\text{ bits}}$ (dramatically boosts resistance to brute force).
- **Drawback:** Processing three DES rounds per block makes 3DES **3 times slower** than single DES.

3.3 Advanced Encryption Standard (AES / Rijndael)

Selected by NIST in 2001, AES (Rijndael) is the current gold standard for symmetric encryption. It uses a fixed **128-bit block size** (16 bytes formatted as a 4×4 byte matrix called the *State*) and supports three key lengths:

AES Variant	Key Length (Bits)	Number of Rounds	Security & Application Level
AES-128	128 bits	10 rounds	Standard commercial encryption grade.
AES-192	192 bits	12 rounds	Enhanced security grade.
AES-256	256 bits	14 rounds	Top-secret military / government grade.

The Four Core Transformations per AES Round:

1. **SubBytes (Substitution):** A non-linear byte-by-byte replacement using an S-Box table (provides *Confusion*).
2. **ShiftRows (Transposition):** Cyclically shifts bytes in each row of the 4×4 matrix by offsets 0, 1, 2, and 3 (provides *Diffusion*).
3. **MixColumns (Mixing):** Combines bytes within each column using Galois Field $GF(2^8)$ matrix multiplication (omitted in the final round).
4. **AddRoundKey (Key Addition):** XORs the 4×4 State matrix directly with the 128-bit subkey derived from the main key.

EXAM POINT: DES vs 3DES vs AES SUMMARY

Summary Comparison of Private-Key Algorithms:
- **DES:** Block = 64 bits | Key = 56 bits | Structure = 16-round Feistel | Status = *Broken (Key too short)*
- **3DES:** Block = 64 bits | Key = 168 bits | Structure = EDE Mode | Status = *Legacy / Slow*
- **AES:** Block = 128 bits | Key = 128/192/256 bits | Structure = Parallel State Rounds | Status = *Active Gold Standard*

MODULE 4: LECTURE 29 — PUBLIC-KEY CRYPTOGRAPHY & THE RSA ALGORITHM

4.1 The Asymmetric Cryptography Paradigm Shift

1. Simple Explanation: In public-key cryptography, every person has a pair of two mathematical keys: a **Public Key** (which you publish openly like a mail slot where anyone can drop a letter) and a **Private Key** (which only you possess to unlock your mailbox).

2. Key Scaling Advantage: For N parties, public-key systems require only **$2N$ total keys** (1 public + 1 private key per user).

FORMULA SHEET & WORKED EXAMPLE: PUBLIC-KEY SCALABILITY

Asymmetric Key Requirement Formula:
$$\text{Total Keys Required} = 2N$$
• **Numerical Example (NPTEL Assignment 6, Q9):** If 150 parties want to exchange secure messages using public-key encryption (RSA):
$$\text{Total Keys} = 2 \times 150 = \mathbf{300 \text{ distinct keys.}}$$
• **Comparison:** Contrast this with symmetric encryption which would require $\frac{150 \times 149}{2} = 11,175$ keys!

4.2 Dual Applications: Confidentiality vs. Digital Signatures

The key pair relationship provides two complementary operational modes:

- **1. Confidential Encryption (NPTEL Assignment 6, Q7):** If Sender A wants to send a confidential message to Receiver B, Sender A encrypts the message using **Receiver B's Public Key** (e_B, n_B). Only Receiver B can decrypt it using **Receiver B's Private Key** (d_B, n_B).
- **2. Digital Authentication / Signatures:** If Sender A wants to prove message origin, Sender A encrypts/signs the message using **Sender A's Private Key**. Anyone can verify the signature using **Sender A's Public Key**.

4.3 The RSA Algorithm Step-by-Step

Named after Rivest, Shamir, and Adleman, RSA's security relies on the **Prime Factoring Problem** — given $n = p \cdot q$, finding p and q for large 2048-bit numbers is computationally infeasible.

Step-by-Step RSA Key Generation & Encryption Mechanics:

1. **Select Primes:** Choose two large prime numbers p and q.
2. **Compute Modulus n:** $n = p \times q$
3. **Compute Euler Totient $\phi(n)$:** $\phi(n) = (p - 1)(q - 1)$
4. **Select Public Exponent e:** Choose integer e such that $1 < e < \phi(n)$ and $\gcd(e, \phi(n)) = 1$.
5. **Compute Private Exponent d:** Calculate d as the multiplicative inverse of $e \bmod \phi(n)$:
   $$d \cdot e \equiv 1 \pmod{\phi(n)}$$
6. **Form Keys:** Public Key = (e, n); Private Key = (d, n).
7. **Encryption Formula:** $C = M^e \bmod n$ (where plaintext $0 \le M < n$).
8. **Decryption Formula:** $M = C^d \bmod n$

SOLVED LECTURE EXAMPLE: COMPLETE RSA COMPUTATION

Complete Worked Lecture Numerical:
• *Given Primes:* $p = 7$, $q = 17$
• *Step 1 (Modulus):* $n = 7 \times 17 = 119$
• *Step 2 (Totient):* $\phi(n) = (7 - 1)(17 - 1) = 6 \times 16 = 96$
• *Step 3 (Public Exponent):* Choose $e = 5$ (Valid since $\gcd(5, 96) = 1$).
• *Step 4 (Private Exponent):* Find d such that $5d \equiv 1 \pmod{96} \implies 5d = 385 = (4 \times 96 + 1) \implies d = 77$.
• **Keys Generated:** Public Key = $(5, 119)$; Private Key = $(77, 119)$.
• **Encrypt Plaintext $M = 19$:**
$$C = 19^5 \bmod 119 = 2,476,099 \bmod 119 = \mathbf{66}$$
• **Decrypt Ciphertext $C = 66$:**
$$M = 66^{77} \bmod 119 = \mathbf{19 \text{ (Plaintext Recovered!)}}$$

4.4 Hybrid Cryptosystems & Quantum Threats

• **Hybrid Cryptosystems:** Public-key algorithms are thousands of times slower than symmetric algorithms due to large modular exponentiations. In practice (e.g., HTTPS/TLS), RSA is used only for secure **Key Exchange** (encrypting a small symmetric key), while fast **AES** handles bulk payload encryption.

• **Quantum Computing Threat:** Large quantum computers running **Shor's Algorithm** can factor large integers n in polynomial time, threatening RSA. This necessitates transition to **Post-Quantum Cryptography (PQC)**.

MODULE 5: LECTURE 30 — DIFFIE-HELLMAN KEY EXCHANGE & MESSAGE AUTHENTICATION

5.1 Diffie-Hellman Key Exchange Protocol Mechanics

1. Simple Explanation: Diffie-Hellman allows two strangers across an insecure network to agree upon a shared secret key without ever directly sending the secret key itself over the wire.

2. Important Distinction: Diffie-Hellman is strictly a *Key Agreement Protocol* — it does NOT encrypt or decrypt general message payloads like RSA or AES.

3. Underlying Hard Mathematical Problem: Security relies on the **Discrete Logarithm Problem** — given public prime n, generator g, and public value $X = g^x \bmod n$, calculating the private exponent x is computationally intractable.

4. Step-by-Step Protocol Execution:

• *Step 1 (Public Parameter Setup):* Parties A and B publicly agree on prime n and generator g ($g < n$).

• *Step 2 (Private Exponent Selection):* Party A selects secret x; Party B selects secret y.

• *Step 3 (Public Value Exchange):*
Party A computes $X = g^x \bmod n$ and transmits X to B.
Party B computes $Y = g^y \bmod n$ and transmits Y to A.

• *Step 4 (Shared Key Calculation):*
Party A computes $K_A = Y^x \bmod n = (g^y)^x \bmod n = g^{xy} \bmod n$.
Party B computes $K_B = X^y \bmod n = (g^x)^y \bmod n = g^{xy} \bmod n$.
Since $K_A = K_B = K$, both parties possess the exact same symmetric key K!

SOLVED LECTURE EXAMPLE: DIFFIE-HELLMAN COMPUTATION

Worked Lecture Numerical Example:
• *Public Parameters:* Prime $n = 11$, Generator $g = 2$.
• *Private Exponents:* Party A chooses $x = 3$; Party B chooses $y = 4$.
• *Public Values Computed & Exchanged:*
Party A: $X = 2^3 \bmod 11 = 8 \bmod 11 = \mathbf{8}$ (Sent to B).
Party B: $Y = 2^4 \bmod 11 = 16 \bmod 11 = \mathbf{5}$ (Sent to A).
• *Shared Key Calculation:*
Party A computes: $K_A = 5^3 \bmod 11 = 125 \bmod 11 = \mathbf{4}$.
Party B computes: $K_B = 8^4 \bmod 11 = 4096 \bmod 11 = \mathbf{4}$.
Result: Both parties establish the shared key $K = 4$ securely!

5.2 Man-in-the-Middle (MitM) Attack on Diffie-Hellman

The MitM Vulnerability: Unauthenticated Diffie-Hellman is vulnerable to an active MitM attack because neither party authenticates the identity of the other.

- *Attack Scenario:* Intruder I intercepts A's public value X and substitutes $X_I = g^{x_I} \bmod n$ to B. Simultaneously, I intercepts B's public value Y and substitutes $Y_I = g^{y_I} \bmod n$ to A.
- *Outcome:* A establishes a secret key K_{AI} with the intruder, and B establishes K_{BI} with the intruder. I transparently decrypts, modifies, and re-encrypts all communications.
- *Countermeasure:* Combine Diffie-Hellman with digital signatures or public key certificates (e.g., Signed Diffie-Hellman).

5.3 Message Authentication & Cryptographic Hash Functions

• **Message Authentication Code (MAC):** A small block of data generated using a shared secret key k and message M ($MAC = F(M, k)$). Appended to the message so the receiver verifies origin and integrity.

• **Cryptographic Hash Functions:** One-way mathematical functions that map arbitrary-length inputs into a fixed-size digest (e.g., MD5 128-bit, SHA-1 160-bit, SHA-256 256-bit).

EXAM POINT: HASH FUNCTION PROPERTIES & MYTHS

Essential Properties of Cryptographic Hash Functions (NPTEL Assignment 6, Q4):
1. **Pre-image Resistance (One-Way Property):** Given a hash digest H, it is computationally impossible to reverse the function to retrieve the original message M.
2. **Second Pre-image Resistance:** Given M_1, it is impossible to find another $M_2 \neq M_1$ such that $Hash(M_1) = Hash(M_2)$.
3. **Collision Resistance:** It is computationally impossible to find any two distinct inputs M_1, M_2 that produce $Hash(M_1) = Hash(M_2)$.
4. **Avalanche Effect:** Changing a single bit in the input causes a drastic, randomized change in the output hash digest.
• **Common Exam Mistake:** Claiming that a hash function can be reversed to recover plaintext is **FALSE**!

MODULE 6: WEEK 6 REVISION & NPTEL EXAM PREPARATION PAGES

6.1 Quick Week 6 Core Revision Summary

- **Security Attack Categories:** Interruption (Availability), Interception (Confidentiality), Modification (Integrity), Fabrication (Authenticity).
- **Passive vs. Active:** Passive attacks (eavesdropping/traffic analysis) leave no trace and are hard to detect; Active attacks (masquerade, replay, DoS) alter system state and are easier to detect.

- **Key Distribution Formulas:** Symmetric encryption requires $N(N-1)/2$ keys; Asymmetric encryption requires $2N$ keys.
- **Classical Ciphers:** Caesar Cipher (vulnerable to brute force; 25 shifts); Mono-alphabetic Cipher (vulnerable to frequency analysis); Transposition Cipher (rearranges letter positions).
- **Practical Private-Key Ciphers:** DES (64-bit block, 56-bit key, 16-round Feistel); 3DES (168-bit key, EDE mode, 3x slower); AES (128-bit block, 128/192/256-bit key, SubBytes/ShiftRows/MixColumns/AddRoundKey).
- **RSA Algorithm:** Asymmetric cipher based on prime factoring ($n = p \cdot q, \phi(n) = (p-1)(q-1), d e \equiv 1 \bmod \phi(n), C = M^e \bmod n, M = C^d \bmod n$).
- **Diffie-Hellman Key Exchange:** Establishes shared symmetric key over insecure channel based on Discrete Logarithm Problem ($X = g^x \bmod n, Y = g^y \bmod n, K = g^{xy} \bmod n$); vulnerable to MitM without authentication.
- **Hash Functions:** One-way, collision-resistant functions (MD5, SHA-1, SHA-256); CANNOT be reversed!

6.2 Master Glossary & Core Terminology

Technical Term	Technical Definition	Exam / Security Significance
Plaintext (P)	Original, readable unencrypted data.	Input to encryption algorithm; must be protected.
Ciphertext (C)	Encrypted, unreadable garbled data output.	Transmitted across untrusted channels.
Cryptanalysis	Attempting to decode ciphertext without possessing the secret key.	Evaluates security strength against mathematical attacks.
Masquerade Attack	An attacker pretends to be a legitimate user or entity.	Active attack targeting Authenticity.
Traffic Analysis	Observing communication patterns, frequency, and volume.	Passive attack on Confidentiality; hard to detect.
Feistel Structure	Cipher design splitting block into 2 halves with identical round logic.	Used in DES; simplifies hardware/software implementation.
SubBytes	Non-linear byte substitution transformation in AES.	Provides Confusion between key and ciphertext.
ShiftRows & MixColumns	Cyclic row shifting and column mixing in AES.	Provides Diffusion, spreading bit influence across the block.
Euler Totient $\phi(n)$	Counts positive integers up to n relatively prime to n.	Core component in RSA key generation ($\phi(n)=(p-1)(q-1)$).
Discrete Logarithm	Hard problem of finding x given $g, n,$ and $g^x \bmod n$.	Forms security basis of Diffie-Hellman key exchange.

6.3 Key Concept Master Comparisons

Comparison Category	Concept A	Concept B	Key Distinguishing Factor
Cryptographic Paradigm	**Symmetric Cryptography** Same secret key for encrypt & decrypt.	**Asymmetric Cryptography** Public/Private key pair.	Symmetric is 1000x faster; Asymmetric solves key distribution ($2N$ vs $N(N-1)/2$).
Attack Nature	**Passive Attack** Eavesdropping/traffic analysis.	**Active Attack** Modification, masquerade, DoS.	Passive alters no data (hard to detect); Active alters system state (easier to detect).
Classical Ciphers	**Caesar Cipher** Fixed letter shift (k).	**Mono-alphabetic Cipher** Arbitrary letter substitution.	Caesar broken by brute force (25 keys); Mono-alphabetic broken by frequency analysis.
Processing Architecture	**Stream Cipher** Processes bit-by-bit continuously.	**Block Cipher** Processes fixed-size blocks (e.g. 128-bit).	Stream ciphers suit live audio/video; Block ciphers suit bulk stored data.
Private-Key Standards	**DES Algorithm** 64-bit block, 56-bit key.	**AES Algorithm** 128-bit block, 128/192/256 key.	DES is broken (key too short); AES is active gold standard.
Public-Key Protocols	**RSA Algorithm** Encryption, Signatures & Key Exchange.	**Diffie-Hellman Protocol** Key Agreement Only.	RSA encrypts general payloads; DH only establishes a shared secret key.

6.4 Master Formula Sheet

Formula Name	Mathematical Equation	Variables Defined	Application Purpose
Symmetric Keys	$$\text{Keys} = \frac{N(N-1)}{2}$$	$N = \text{number of parties}$	Calculates total symmetric keys for N parties.
Asymmetric Keys	$$\text{Keys} = 2N$$	$N = \text{number of parties}$	Calculates total public/private keys for N parties.
Caesar Encryption	$$C = (P + k - 1) \bmod 26 + 1$$	$P = \text{letter}, k = \text{shift}$	Encrypts letter in Caesar cipher.
Caesar Decryption	$$P = (C - k + 25) \bmod 26 + 1$$	$C = \text{letter}, k = \text{shift}$	Decrypts letter in Caesar cipher.
RSA Modulus & Totient	$$n = p \cdot q, \quad \phi(n) = (p-1)(q-1)$$	$p, q = \text{large prime numbers}$	RSA key generation setup.
RSA Key Relation	$$d \cdot e \equiv 1 \pmod{\phi(n)}$$	$e = \text{public exp}, d = \text{private exp}$	Computes private exponent d.
RSA Encrypt / Decrypt	$$C = M^e \bmod n, \quad M = C^d \bmod n$$	$M = \text{plaintext}, C = \text{ciphertext}$	RSA transformation operations.
Diffie-Hellman Key	$$K = (g^y)^x \bmod n = g^{xy} \bmod n$$	$n, g = \text{public}, x, y = \text{private}$	Calculates shared symmetric key.

6.5 Important Solved Examples

SOLVED EXAMPLE 1: RSA Key Generation & Encryption

Problem: Given primes $p = 3, q = 11$, public exponent $e = 7$, encrypt plaintext $M = 2$.

Solution:

1. $n = p \cdot q = 3 \times 11 = 33$.
2. $\phi(n) = (3 - 1)(11 - 1) = 2 \times 10 = 20$.
3. Find d such that $7d \equiv 1 \pmod{20} \implies 7d = 21 \implies d = 3$.
4. Public Key = $(7, 33)$; Private Key = $(3, 33)$.
5. Encrypt $M = 2$: $C = M^e \bmod n = 2^7 \bmod 33 = 128 \bmod 33 = \mathbf{29}$.

SOLVED EXAMPLE 2: Diffie-Hellman Shared Key Computation

Problem: Public parameters $n = 23, g = 5$. Alice private $x = 6$, Bob private $y = 15$. Find shared key K.

Solution:

1. Alice public $X = g^x \bmod n = 5^6 \bmod 23 = 15,625 \bmod 23 = 8$.
2. Bob public $Y = g^y \bmod n = 5^{15} \bmod 23 = 19$.
3. Shared key $K = Y^x \bmod 23 = 19^6 \bmod 23 = 47,045,881 \bmod 23 = \mathbf{2}$.

6.6 NPTEL Exam Practice MCQs (With Solutions)

Q1: Which of the following attack types involves an attacker pretending to be a legitimate entity to gain unauthorized access?

a. Replay attack | b. Masquerade attack | c. Modification attack | d. Denial of service attack | e. Traffic analysis

Correct Answer: b

Explanation: In a masquerade attack, an attacker pretends to be a legitimate entity to gain unauthorized access.

Q2: Encrypt the plaintext 'SECURE' using a substitution cipher where each letter is replaced by its k-th next letter (k = 3).

a. VHFXUH | b. VFEXRU | c. UHFXVH | d. VHFXRV | e. None of these

Correct Answer: a

Explanation: Shifting each letter by 3 positions: S->V, E->H, C->F, U->X, R->U, E->H gives 'VHFXUH'.

Q3: Consider a mono-alphabetic cipher with key mapping (A B C D E ...) mapped to (Q W E R T ...). What is the encrypted form of 'CODE'?

a. EGRT | b. EGDT | c. EODT | d. EGDR | e. None of these

Correct Answer: a

Explanation: Using the given mapping key: C->E, O->G, D->R, E->T. Thus, 'CODE' encrypts to 'EGRT'.

Q4: Which of the following statement(s) is/are FALSE regarding hashing algorithms used for message integrity?

a. Resistant to collisions | b. Deterministic output | c. A hash function can be reversed to retrieve original message | d. Exhibits avalanche effect

Correct Answer: c

Explanation: Cryptographic hash functions are one-way functions and CANNOT be reversed to retrieve original plaintext.

Q5: Decrypt the ciphertext 'KHOOR' encrypted using a substitution cipher with a shift of k = 3.

a. HELLO | b. WORLD | c. HELPS | d. HFLLO | e. None of these

Correct Answer: a

Explanation: Shifting back by 3 positions: K->H, H->E, O->L, O->L, R->O gives 'HELLO'.

Q6: A DoS attack that makes a system or service inaccessible to legitimate users falls under which category of security attack?

a. Interruption | b. Interception | c. Modification | d. Fabrication | e. Replay

Correct Answer: a

Explanation: A Denial of Service (DoS) attack prevents access to services, which is classified as an Interruption attack on Availability.

Q7: For encryption using public-key cryptography, if sender A wants to send a confidential message to receiver B, which key is used?

a. A's private key | b. A's public key | c. B's public key | d. B's private key

Correct Answer: c

Explanation: For confidentiality, the sender encrypts using the receiver's public key so only the receiver can decrypt with their private key.

Q8: What is the block size (in bits) used for plaintext blocks in the Data Encryption Standard (DES) algorithm?

a. 32 bits | b. 56 bits | c. 64 bits | d. 128 bits

Correct Answer: c

Explanation: DES processes plaintext in fixed 64-bit blocks using a 56-bit effective key.

Q9: 150 parties want to exchange messages securely using a public key encryption technique like RSA. How many distinct key values are required?

a. 150 | b. 300 | c. 11,175 | d. 22,350

Correct Answer: b

*Explanation: Public-key cryptography requires 2 keys per party (public + private). Total keys = 2 * 150 = 300.*

Q10: 40 parties want to exchange messages securely using a symmetric key encryption algorithm. How many distinct keys will be required?

a. 40 | b. 80 | c. 780 | d. 1560

Correct Answer: c

*Explanation: Symmetric key requirement for N parties is N(N-1)/2. For N = 40, 40 * 39 / 2 = 780 keys.*

6.7 Oral Viva Questions & Short Answers

Q: 1. What is Kerckhoffs's Principle?

A: Security must depend solely on key secrecy, not on keeping the algorithm secret.

Q: 2. What are the 4 categories of security attacks?

A: Interruption (Availability), Interception (Confidentiality), Modification (Integrity), Fabrication (Authenticity).

Q: 3. Why are passive attacks harder to detect than active attacks?

A: Passive attacks leave no trace and alter no data, whereas active attacks alter system state.

Q: 4. How many keys are needed for N parties in symmetric vs. asymmetric systems?

A: Symmetric: N(N-1)/2 keys. Asymmetric: 2N keys.

Q: 5. What is the main weakness of the Caesar Cipher?

A: Very small key space (only 25 possible shifts) making it vulnerable to brute force.

Q: 6. How is a Mono-alphabetic cipher broken?

A: Using Letter Frequency Analysis (matching ciphertext letter frequencies to natural language statistics).

Q: 7. What is the block size and key size of DES?

A: Block size = 64 bits; Effective key size = 56 bits.

Q: 8. Why does AES use non-Feistel structure?

A: To process all 128 bits of the state matrix in parallel during each round for high speed.

Q: 9. What hard mathematical problem secures RSA?

A: The Integer Prime Factoring Problem (factoring n = p * q).

Q: 10. What hard mathematical problem secures Diffie-Hellman?

A: The Discrete Logarithm Problem.

Q: 11. Why is Diffie-Hellman vulnerable to Man-in-the-Middle attacks?

A: Because basic Diffie-Hellman does not authenticate the identities of the communicating parties.

Q: 12. What is a Cryptographic Hash Function?

A: A one-way, collision-resistant function that maps arbitrary inputs to a fixed-size digest.

6.8 Final Revision Checklist

[ ] Understand Interruption, Interception, Modification, and Fabrication attack categories.

[ ] Differentiate Passive attacks (eavesdropping) from Active attacks (masquerade, DoS).

[ ] Master symmetric key calculation $N(N-1)/2$ vs asymmetric $2N$.

[ ] Perform Caesar cipher encryption/decryption shift math.

[ ] Explain why Mono-alphabetic ciphers fail against Frequency Analysis.

[ ] Memorize DES parameters (64-bit block, 56-bit key, 16 Feistel rounds).

[ ] Explain AES round transformations: SubBytes, ShiftRows, MixColumns, AddRoundKey.

[ ] Compute RSA steps: $n = p \cdot q, \phi(n) = (p-1)(q-1), d e \equiv 1 \bmod \phi(n), C = M^e \bmod n, M = C^d \bmod n$.

[ ] Compute Diffie-Hellman steps: $X = g^x \bmod n, Y = g^y \bmod n, K = g^{xy} \bmod n$.

[ ] Memorize properties of cryptographic hash functions (one-way, collision resistant).